

Jegadeesh D

✉ jegadeeshmsd@gmail.com ☎ 9585164547 🌐 jegadeesh17 🌐 jegadeesh17

PROFILE

Quantitative Developer with a strong foundation in statistical modeling and Python. Architect of an institutional-grade algorithmic trading pipeline featuring Combinatorial Purged Cross-Validation (CPCV), multi-factor alpha models, and derivatives pricing engines

CORE SKILLS

Programming Languages: Python (Intermediate), SQL (PostgreSQL, TimescaleDB)

Quantitative Libraries: Pandas, NumPy, SciPy, Statsmodels, Scikit-Learn

Financial Engineering: Algorithmic Trading, Options Pricing (Black-Scholes, Greeks), Volatility Modeling (GARCH), Transaction Cost Analysis (TCA), Implied Volatility (Newton-Raphson), Portfolio Optimization (Kelly Criterion)

Statistical Modeling: Time Series Analysis, Combinatorial Purged Cross-Validation (CPCV), Deflated Sharpe Ratio, Statistical Arbitrage, Principal Component Analysis (PCA)

Tools & Infrastructure: Git, REST APIs, JSON, Streamlit

PROJECTS

Quandao - Multi-Asset Quantitative Research & Execution Platform

Architected a multi-asset algorithmic trading system for NSE Equities, Forex, and Commodities (Gold), handling alpha generation, rigorous statistical validation, and simulated options execution

Project 1: Multi-Factor Alpha Research Engine

- Engineered a vectorized multi-factor alpha engine (Pandas, PostgreSQL) processing 1-minute OHLCV multi-asset data (2017-present) to compute cross-sectional signals (Momentum, Amihud Illiquidity).
- Implemented dynamic factor weighting via 63-day Spearman Rank Information Coefficients (IC), applying sector neutralization to isolate idiosyncratic alpha.
- Developed a GARCH(1,1) conditional volatility model forecasting market regimes, automating position-sizing to maintain constant portfolio risk.

Project 2: Combinatorial Purged Backtesting & Risk Framework

- Built a rigorous backtester utilizing Combinatorial Purged Cross-Validation (CPCV) with test-set embargoes, strictly eliminating temporal data leakage.
- Programmed the Deflated Sharpe Ratio (DSR) with Euler-Mascheroni corrections, penalizing multiple-testing bias to quantify Probability of Backtest Overfitting (PBO).
- Developed a dynamic risk manager integrating drawdown circuit breakers, net-delta caps, and Kelly Criterion sizing to optimize compounded growth.

Project 3: Low-Latency Options Execution & TCA Simulator

- Built a vectorized Black-Scholes pricing engine computing continuous-time Greeks (Δ , Γ , Θ , ν , ρ) for live risk monitoring and delta-targeted strike selection.
- Programmed a Newton-Raphson root-finding algorithm (Brent's method fallback) to reverse-engineer Implied Volatility from option premiums in <10 iterations.
- Developed a Transaction Cost Analysis (TCA) simulator, isolating Net Alpha by deducting NSE statutory fees, Bid-Ask slippage, and Almgren-Chriss market impact.

EDUCATION

MTech (Integrated) in Software Engineering

Vellore Institute Of Technology Chennai

Current CGPA-7.89/10

01/07/2020 –

01/07/2025

CERTIFICATIONS

Financial Markets, Yale University | Microsoft Certified: Security, Compliance, and Identity Fundamentals |

Google Project Management | Financial Planning for Young Adults